Data Analysis from TP

Multi-Agent Systems for Smart Machining - ECIU University

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Introduction

The machining process under analysis consists of three main operations: surface milling, drilling, and pocket milling. This document compiles the analyses of the various machining operations and ends with a dedicated section that provides a comparison of the corresponding results.

By examining the power profile over time, it is possible to identify the different phases of the machining process. The Figure 1 shows the total power P_{tot} measured during the cutting session, together with the cutting power component P_c, computed as $P_c = P_{tot} - P_{nc}$, where P_{nc} represents the no-load power measured during the NON-cutting phase.

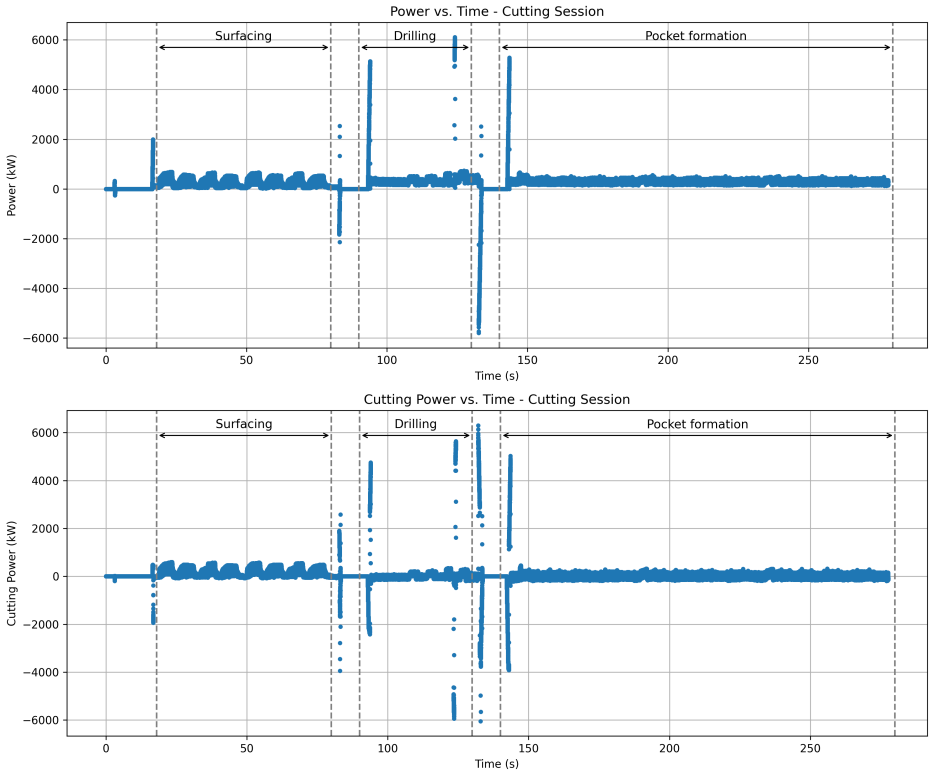


Figure 1: Total power P_{tot} and cutting power P_c during the surface milling operation.

Ignoring the instantaneous power peaks, it can be observed that the cutting power frequently becomes negative, which is physically impossible, as it would imply that the cutting process generates energy. This

behaviour suggests an inconsistency between the cutting and NON-cutting sessions. For this reason, the analysis will be carried out using the total power measured during the cutting session. This choice remains fully representative, since including the no-load power would simply result in a constant offset of the power values.

Each operation was analyzed by one ECIU student:

- Paolo Rossi: surface milling;
- Chloé Fuchs: drilling;
- Daniele Trisotto: pocket milling;

Qua aggiungero le nostre analisi del cazzo

Comparison between the different operations

When comparing the three machining operations, both the power and energy profiles clearly reflect the underlying process kinematics. Despite the different spindle speeds imposed by the varying tool diameters, the instantaneous power traces consistently reveal the structure of each operation. In drilling, the step-like pattern corresponds directly to the sequence of four holes, with each plateau representing one drilling event. Conversely, in surface milling and pocket formation, the oscillatory behaviour of the power signal is associated with the continuous changes in tool engagement caused by the toolpath direction reversals on the workplane.

The average power per revolution falls within a similar range for all operations—approximately 200 to 500 W/rev. This is coherent with the fact that the cutting speed is kept constant across the three processes and the work material does not change.

Regarding energy, the profiles exhibit the same underlying behaviour observed in the power signals, since energy is obtained by integrating power over time. When analysing the per-revolution values, surface milling shows the highest energy consumption, with approximately 30 J per revolution (or slightly below), whereas drilling and pocket formation remain around 4 J per revolution.